Capabilities and Limitations of Composition-Only Machine Learning for Predicting the Thermodynamic Stability of Multicomponent Zn-, In-, and Sn-Based Oxides

Interactive walk-through of the complete pipeline (`seed = 42`). Each section calls the corresponding script in `scripts/` , ensuring identical results between the notebook and command-line execution.

Requirements: `environment.yml` or `requirements.txt` . Place the archived raw dataset in `data/raw/mp_doped_oxides_raw.csv` (or set `MP_API_KEY` and set `REFRESH_RAW = True` below).

In [1]:
```
import subprocess, sys
from pathlib import Path
import pandas as pd
from IPython.display import Image, display

# Locate the project root (folder containing environment.yml)
ROOT = next(p for p in [Path.cwd(), *Path.cwd().parents] if (p / "environment.ym
REFRESH_RAW = False   # Set True to re-query the Materials Project API

def run(script):
    """Run one pipeline script (same env as run_all.py), stream output live, log
    import os
    print(f"\n>>> {script}")
    env = os.environ.copy()
    env.update(PYTHONUTF8="1", PYTHONIOENCODING="utf-8", JOBLIB_MULTIPROCESSING=
    log_dir = ROOT / "logs" / "steps"; log_dir.mkdir(parents=True, exist_ok=True
    log = log_dir / f"nb_{Path(script).stem}.log"
    lines = []
    with open(log, "w", encoding="utf-8") as fh:
        p = subprocess.Popen([sys.executable, str(ROOT / "scripts" / script)], c
                             stdout=subprocess.PIPE, stderr=subprocess.STDOUT,
                             text=True, encoding="utf-8", errors="replace", bufs
        for line in p.stdout:
            print(line, end=""); fh.write(line); lines.append(line)
        p.wait()
    if p.returncode:
        print("".join(lines[-25:]))
        raise RuntimeError(f"{script} failed (code {p.returncode}) - log: {log}"

def show(*patterns):
    for pat in patterns:
        for f in sorted(ROOT.glob(pat)):
            print(f.relative_to(ROOT)); display(Image(filename=f))

print("Project root:", ROOT)
```

```
Project root: d:\tco_stability_repo
```

Phase 1 - Data Acquisition, Cleaning, and Exploration (Sections 2.1-2.2, 3.1)

In [2]:
```
run("01_check_env.py")
if REFRESH_RAW or not (ROOT / "data/raw/mp_doped_oxides_raw.csv").exists():
    run("02_test_api.py")
    run("03_query_materials.py")
for s in ["04_clean_data.py", "05_count_host_structures.py", "06_sensitivity_ef.
          "07_plot_data_exploration.py", "08_plot_ehull_by_host.py"]:
    run(s)
```

```
>>> 01_check_env.py
Python: 3.15.0 | packaged by Anaconda, Inc. | (main, Oct 13 2026, 19:24:54) [MSC
v.1942 64 bit (AMD64)]
  ✓ pymatgen: installed (no __version__)
  ✓ matminer: 0.10.1
  ✓ sklearn: 1.9.1
  ✓ pandas: 2.3.3
  ✓ numpy: 2.5.4
  ✓ matplotlib: 3.11.2
  ✓ seaborn: 0.13.2
  ✓ mp_api: installed (no __version__)
  ✓ tqdm: 4.70.1
  ✓ xgboost: 3.4.1

>>> 04_clean_data.py
Working directory: D:\tco_stability_repo
======================================================================
PHASE 1, STEP 1.4: DATA CLEANING & INITIAL EXPLORATION
======================================================================

[LOAD] Raw data: 7116 rows, 16 columns

--- Step 1: Missing Values ---
  ✓ No missing values found

--- Step 2: Positive Formation Energy Filter ---
  Entries with positive formation energy: 58
  Removed 58 entries. Remaining: 7058

--- Step 3: Energy Above Hull Outlier Filter ---
  Entries with Ehull > 2.0 eV/atom: 2
  Removed 2 entries. Remaining: 7056

--- Step 4: Polymorph Handling ---
  Formulas with multiple entries (polymorphs): 906
  Maximum entries for one formula: 30 (SnO2)
  Kept most stable polymorph per formula: 7056 → 5216 entries

--- Step 5: Data Integrity Checks ---
  ✓ energy_above_hull: all non-negative
  ✓ band_gap: all non-negative
  ✓ density: all positive
  ✓ volume: all positive

======================================================================
CLEANED DATA SUMMARY
======================================================================

Dataset shape: (5216, 16)
Unique formulas: 5216
Number of elements range: 2 - 7

Target variable (energy_above_hull) statistics:
count    5216.000000
mean        0.116611
std         0.166949
min         0.000000
25%         0.019573
50%         0.068186
75%         0.139715
```

```
max         1.578363

Formation energy statistics:
count    5216.000000
mean       -2.153564
std         0.609890
min        -3.567433
25%        -2.581772
50%        -2.279909
75%        -1.795707
max        -0.013487

Band gap distribution:
  Metallic (gap=0):      1030
  Semiconductor (gap>0): 4186

Crystal system distribution:
crystal_system
Triclinic       1604
Monoclinic      1494
Orthorhombic     685
Cubic            548
Trigonal         484
Tetragonal       293
Hexagonal        108

Number of elements distribution:
nelements
2      14
3     781
4    1883
5    1440
6    1080
7      18

Host element distribution:
host_element
Zn    2608
Sn    1659
In     949

Stable (on hull) vs metastable:
is_stable
False    4558
True      658

[SAVE] Cleaned data saved to: data/processed/mp_doped_oxides_cleaned.csv
[SAVE] Cleaning log saved to: data/processed/cleaning_log.txt
[SAVE] Exploration figure saved to: figures/01_data_exploration.png
[SAVE] Host element figure saved to: figures/02_ehull_by_host.png

======================================================================
DATA CLEANING COMPLETE
Raw: 7116 rows → Cleaned: 5216 rows
Rows removed: 1900 (26.7%)
======================================================================

>>> 05_count_host_structures.py
Working directory: D:\tco_stability_repo
======================================================================
```

```
HOST vs MULTICOMPONENT OXIDE AUDIT SUMMARY (Section 3.1)
================================================================================
Source file (read-only)         : data\processed\mp_doped_oxides_cleaned.csv
Rows in cleaned CSV             : 5216
Rows analyzed (featurized set)  : 5207 (exclude_unsupported=True)
--------------------------------------------------------------------------------
Strict dilute-doped hosts       : 1 (0.02%)
  - ZnO-type dilute doped       : 0
  - In2O3-type dilute doped     : 1
  - SnO2-type dilute doped      : 0
Multicomponent oxide phases     : 4987 (95.77%)
Borderline / binary edge cases  : 219 (4.21%)
Total featurized compositions   : 5207
--------------------------------------------------------------------------------
Comparison with relaxed definitions (nelements >= 3, no stoich check):
  - Zn/In/Sn > 90% of non-O     : 1 (0.02%)
  - Any non-O > 90% of non-O    : 19 (0.36%)
--------------------------------------------------------------------------------
Borderline cases exported to    : data\processed\ambiguous_cases.csv
================================================================================

>>> 06_sensitivity_ef.py
Working directory: D:\tco_stability_repo
======================================================================
SENSITIVITY ANALYSIS: IMPACT OF POSITIVE FORMATION ENERGY FILTER (Ef >= 0)
======================================================================
Raw entries with Ef >= 0                : 58
Featurized dataset size (WITH filter)   : 5207
Featurized dataset size (WITHOUT filter): 5223 (+16 unique formulas)
----------------------------------------------------------------------
1. WITH Ef < 0 filter (Standard, n_test=521):
   MAE = 0.0644 eV/atom | RMSE = 0.1094 | R2 = 0.5544
2. WITHOUT filter (Same 521 test formulas, n_test=521):
   MAE = 0.0654 eV/atom | RMSE = 0.1117 | R2 = 0.5354
   -> Absolute MAE difference (same test set): 0.00104 eV/atom
3. WITHOUT filter (Re-shuffled test split, n_test=522):
   MAE = 0.0622 eV/atom | RMSE = 0.1197 | R2 = 0.4364
   -> Absolute MAE difference (re-split)     : 0.00220 eV/atom
======================================================================
Saved: results/sensitivity_ef_results.csv

>>> 07_plot_data_exploration.py
Working directory: D:\tco_stability_repo
Loaded cleaned data: 5216 rows, 16 columns
Figure saved to: figures/phase1/01_data_exploration.png

>>> 08_plot_ehull_by_host.py
Working directory: D:\tco_stability_repo
Loaded cleaned data: 5216 rows, 16 columns

Energy Above Hull statistics by host element:
--------------------------------------------------------
  Zn-oxides: n=2608, mean=0.1257, median=0.0725, std=0.1848
  In-oxides: n= 949, mean=0.0872, median=0.0443, std=0.1404
  Sn-oxides: n=1659, mean=0.1191, median=0.0750, std=0.1486

Figure saved to: figures/phase1/02_ehull_by_host.png
```

In [3]:
```
df = pd.read_csv(ROOT / "data/processed/mp_doped_oxides_cleaned.csv")
print(df.shape)   # expected: (5216, 15)
```

```
df.head()
```

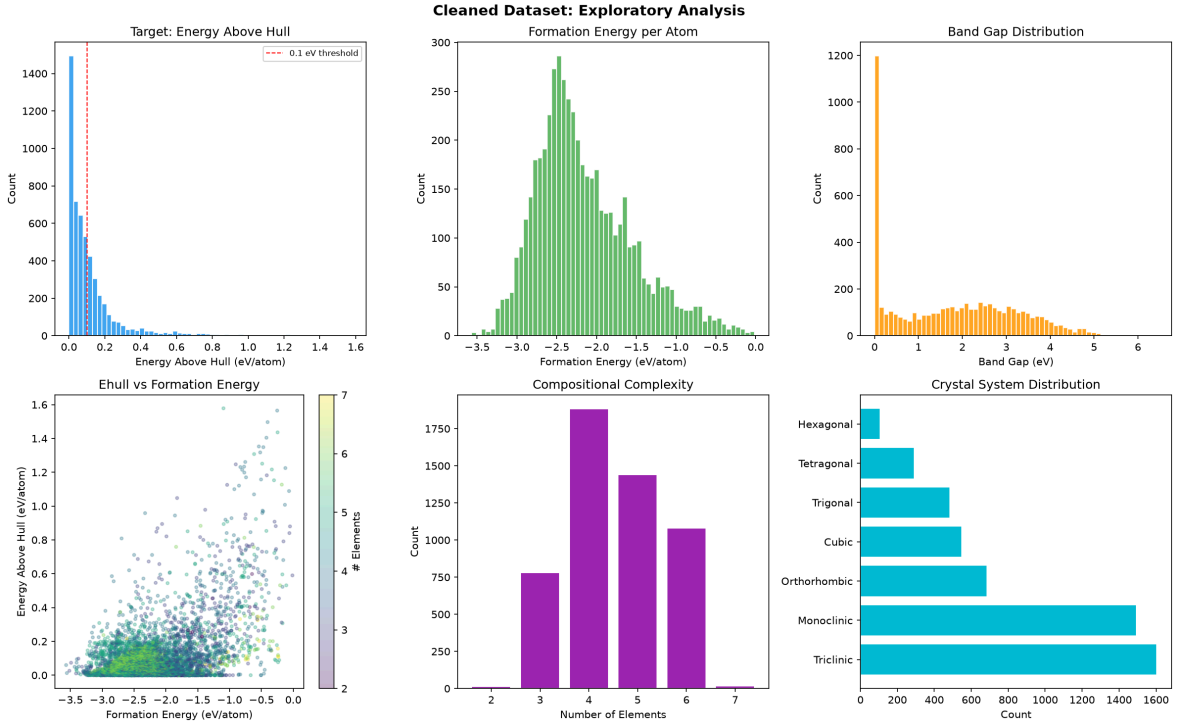
```
(5216, 16)
```

Out[3]:

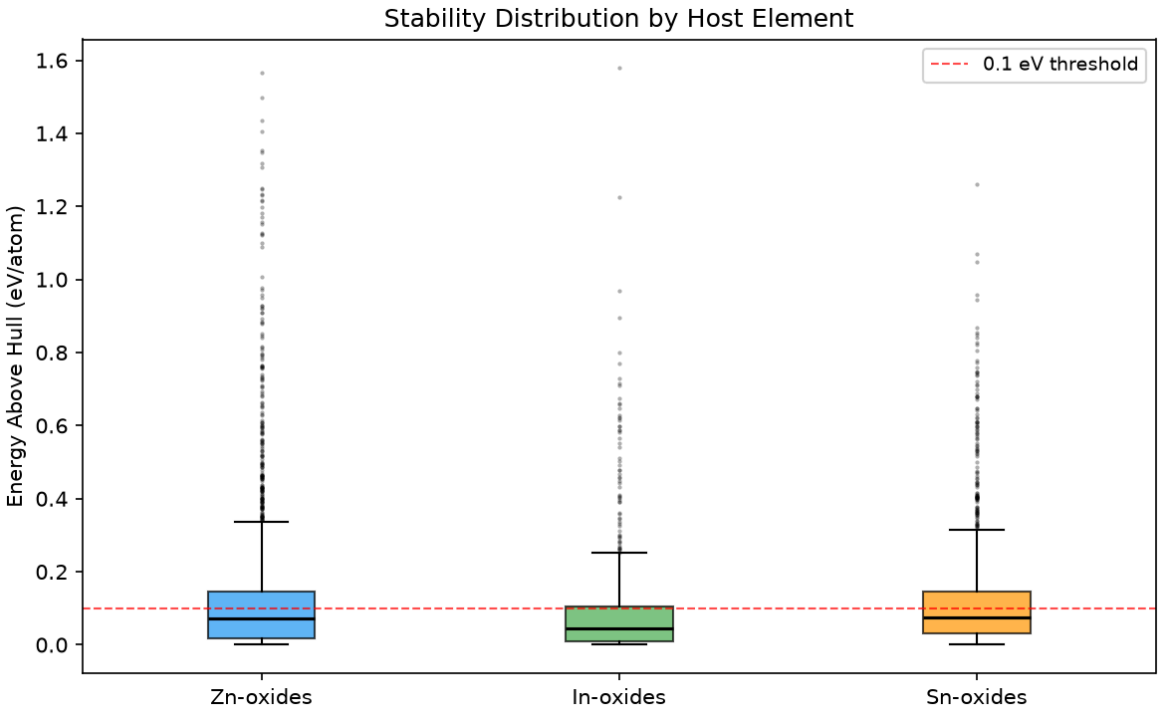
	material_id	formula_pretty	composition	elements	nelements	energy_above_hull	f
0	mp-1007694	SnPO4F	Sn4 P4 O16 F4	F, O, P, Sn	4	0.0	
1	mp-1018031	KInO2	K1 In1 O2	In, K, O	3	0.0	
2	mp-1019525	Ba3ZnB5PO14	Ba6 Zn2 B10 P2 O28	B, Ba, O, P, Zn	5	0.0	
3	mp-1019612	Cs4ZnO3	Cs16 Zn4 O12	Cs, O, Zn	3	0.0	
4	mp-1019786	K3Zn(BO2)5	K12 Zn4 B20 O40	B, K, O, Zn	4	0.0	

In [4]:
```
show("figures/phase1/*.png")
```

```
figures\phase1\01_data_exploration.png
```




```
figures\phase1\02_ehull_by_host.png
```



Phase 2 - Baseline Models, 105 Features (Sections 2.3-2.6, 3.2)

In [5]:
```
run("09_phase2_baseline.py")
pd.read_csv(ROOT / "data/processed/phase2_model_results.csv")
```

```
>>> 09_phase2_baseline.py
Working directory: D:\tco_stability_repo

[STEP 2.1] Loading data and generating CBFVs...
  Loaded: 5216 entries
  After removing unsupported elements: 5207 entries
  Featurizing compositions...
  Generated 105 features for 5207 compositions (6.7s)
  Example features: ['atomic_number_mean', 'atomic_number_std', 'atomic_number_mi
n', 'atomic_number_max', 'atomic_number_range'] ... ['density_elem_range', 'densi
ty_elem_wmean', 'density_elem_wstd']

[STEP 2.2] Splitting data by unique composition...
  Train: 3645 samples (70.0%)
  Val:   1041 samples (20.0%)
  Test:  521 samples (10.0%)

  Data leakage check:
    Train ∩ Val:  0 common formulas ✓
    Train ∩ Test: 0 common formulas ✓
    Val ∩ Test:   0 common formulas ✓
  Splits saved to data/splits/

[STEP 2.3] Scaling features...
  Scaler fit on training set: mean≈0, std≈1
  Train feature range: [-7.15, 12.66]

[STEP 2.4] Training models...
------------------------------------------------------------

  DummyRegressor (baseline - predicts mean):
    MAE  = 0.1041 eV/atom
    RMSE = 0.1757 eV/atom
    R²   = -0.0004

  Training RandomForestRegressor...
    Training time: 3.2s

  RandomForestRegressor results:
                        MAE       RMSE         R²
    Training         0.0366     0.0709     0.8132
    Validation       0.0761     0.1422     0.3444
    Dummy (ref)      0.1041     0.1757    -0.0004

    → RF reduces MAE by 27.0% vs baseline

[VISUALIZATION] Generating parity plot...
  Saved: figures/phase2/03_parity_plot_RF.png
  Saved: figures/phase2/04_residual_plot_RF.png

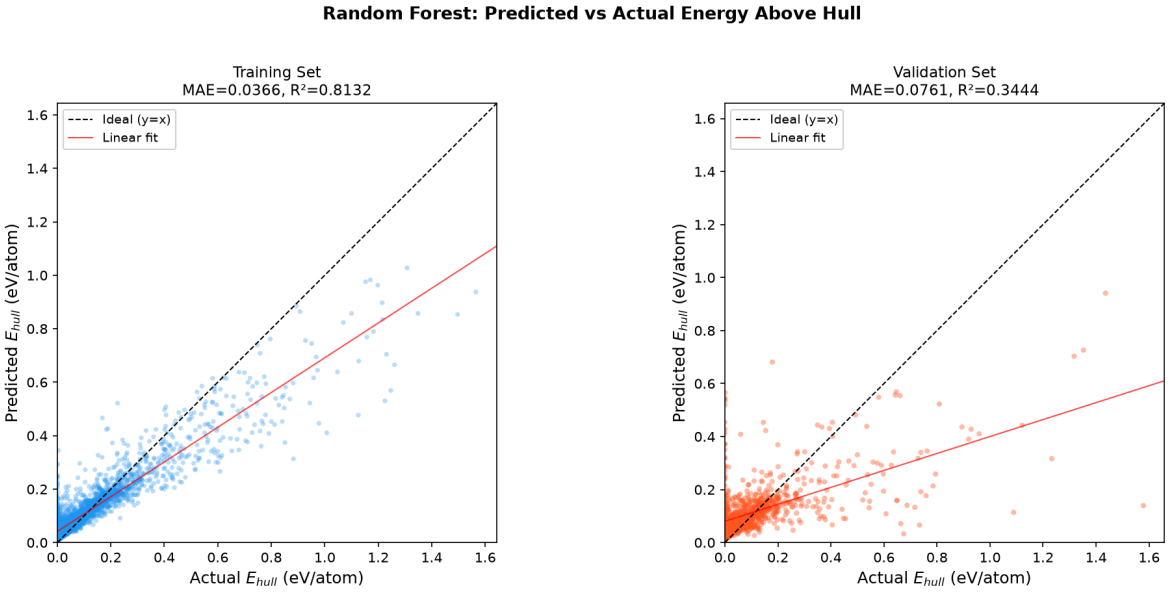
======================================================================
PHASE 2 COMPLETE
======================================================================
  Features: 105 CBFVs from 15 elemental properties × 7 statistics
  Split: 3645 train / 1041 val / 521 test
  Baseline MAE: 0.1041 eV/atom
  RF Val MAE:   0.0761 eV/atom (R²=0.3444)
  Improvement:  27.0% over baseline
======================================================================
```

Out[5]:

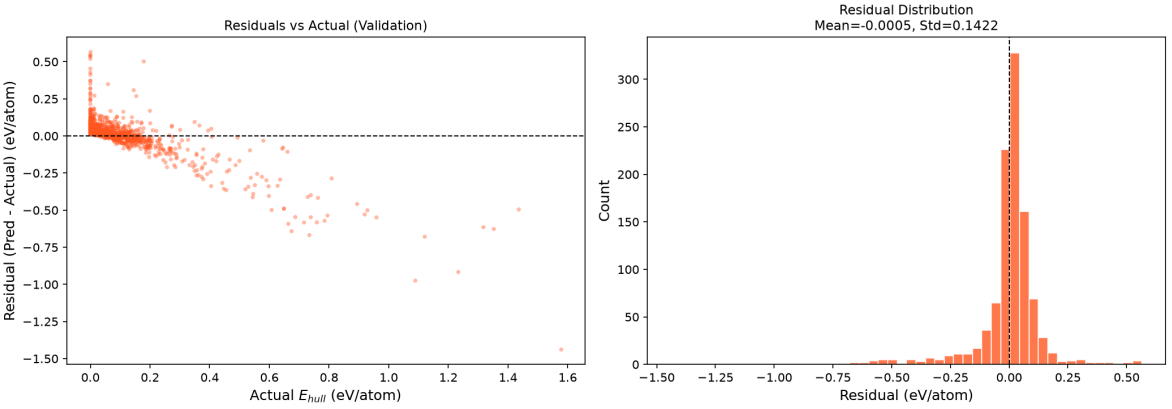
	model	MAE_val	RMSE_val	R2_val	MAE_train	R2_train
0	DummyRegressor	0.104129	0.175663	-0.000358	NaN	NaN
1	RandomForest	0.076055	0.142206	0.344414	0.036629	0.813213

In [6]:
```
for s in ["train", "val", "test"]:
    print(s, len(pd.read_csv(ROOT / f"data/splits/{s}.csv")))   # expected: 3645
show("figures/phase2/*.png")
```

```
train 3645
val 1041
test 521
figures\phase2\03_parity_plot_RF.png
```




```
figures\phase2\04_residual_plot_RF.png
```



Phase 3 - Hyperparameter Tuning and 118-Feature Models (Sections 3.3-3.5)

In [7]:
```
run("10_phase3_tuning_105.py")
run("11_phase3_enhanced_118.py")
pd.read_csv(ROOT / "results/final_metrics_summary.csv")
```

```
>>> 10_phase3_tuning_105.py
Working directory: D:\tco_stability_repo
======================================================================
PHASE 3: MODEL INTERPRETATION, TUNING & FINAL EVALUATION
======================================================================

[SETUP] Reproducing Phase 2 pipeline...
  Data: 3645 train / 1041 val / 521 test
  Features: 105

======================================================================
[STEP 3.1] FEATURE IMPORTANCE ANALYSIS
======================================================================

  Top 20 most important features:
  --------------------------------------------------
   1. melting_point_wmean            0.0393 ████████████████████████
   2. electron_affinity_wmean        0.0321 ███████████████████
   3. electronegativity_wmean        0.0301 ██████████████████
   4. melting_point_wstd             0.0286 █████████████████
   5. ionization_energy_wmean        0.0268 ████████████████
   6. electronegativity_wstd         0.0242 ██████████████
   7. atomic_radius_wmean            0.0223 █████████████
   8. electron_affinity_wstd         0.0218 ████████████
   9. density_elem_wmean             0.0210 ████████████
  10. electronegativity_mean         0.0210 ████████████
  11. n_p_wmean                      0.0201 ███████████
  12. row_wmean                      0.0197 ███████████
  13. ionization_energy_wstd         0.0195 ██████████
  14. n_valence_wmean                0.0181 ██████████
  15. group_wmean                    0.0180 ██████████
  16. ionization_energy_mean         0.0174 ██████████
  17. electron_affinity_std          0.0173 ██████████
  18. atomic_number_wmean            0.0172 ██████████
  19. n_valence_wstd                 0.0171 ██████████
  20. n_p_wstd                       0.0168 ██████████

  Saved: figures/phase3/05_feature_importance_RF.png

======================================================================
[STEP 3.2] PHYSICAL INTERPRETATION
======================================================================

  Key feature → Physical mechanism mapping:
  ================================================================

  1. melting_point_wmean (importance: 0.0393)
     Statistic: wmean
     Physics: Melting point — proxy for bond strength. Elements with high melting
points form stronger bonds, potentially stabilizing ...

  2. electron_affinity_wmean (importance: 0.0321)
     Statistic: wmean
     Physics: Tendency to accept electrons — related to reducibility. Important f
or n-type TCO dopants (e.g., Al in ZnO) where the dop...

  3. electronegativity_wmean (importance: 0.0301)
     Statistic: wmean
     Physics: Bond ionicity — large electronegativity differences between dopant
and host indicate ionic character mismatch, destabili...
```

```
  4. melting_point_wstd (importance: 0.0286)
     Statistic: wstd
     Physics: Melting point — proxy for bond strength. Elements with high melting
points form stronger bonds, potentially stabilizing ...

  5. ionization_energy_wmean (importance: 0.0268)
     Statistic: wmean
     Physics: Ease of electron removal — reflects oxidation state flexibility. Do
pants with very different ionization energies may cre...

  6. electronegativity_wstd (importance: 0.0242)
     Statistic: wstd
     Physics: Bond ionicity — large electronegativity differences between dopant
and host indicate ionic character mismatch, destabili...

  7. atomic_radius_wmean (importance: 0.0223)
     Statistic: wmean
     Physics: Size mismatch — atomic radius differences drive lattice strain when
dopants substitute for host cations. Goldschmidt tol...

  8. electron_affinity_wstd (importance: 0.0218)
     Statistic: wstd
     Physics: Tendency to accept electrons — related to reducibility. Important f
or n-type TCO dopants (e.g., Al in ZnO) where the dop...

  9. density_elem_wmean (importance: 0.0210)
     Statistic: wmean
     Physics: Elemental density — proxy for atomic packing and metallic/covalent
character. Heavy elements pack differently in oxide f...

  10. electronegativity_mean (importance: 0.0210)
     Statistic: mean
     Physics: Bond ionicity — large electronegativity differences between dopant
and host indicate ionic character mismatch, destabili...

  Saved: figures/phase3/06_importance_by_category.png

======================================================================
[STEP 3.3] HYPERPARAMETER TUNING
======================================================================

  Grid search for RandomForest hyperparameters...
  Paper: 'train numerous models with different hyperparameters
  using the same training set, evaluate on the same validation set'

  Total combinations: 108
   n_est  depth  split   leaf |   MAE_tr  MAE_val    R2_tr   R2_val
  ------ ------ ------ ------ | -------- -------- -------- --------
     100     10      2      1 |   0.0636   0.0871   0.6547   0.2893
     100     10      2      2 |   0.0648   0.0866   0.6224   0.2927
     100     10      2      4 |   0.0681   0.0873   0.5525   0.2729
     100   None     10      4 |   0.0500   0.0788   0.6730   0.3203
     200   None     10      4 |   0.0497   0.0787   0.6744   0.3194
     300   None     10      4 |   0.0496   0.0784   0.6741   0.3231

  Grid search completed in 311.3s

  BEST hyperparameters (by validation MAE):
    n_estimators     = 300
```

```
    max_depth        = 30.0
    min_samples_split = 2
    min_samples_leaf  = 1
    Val MAE = 0.0750, Val R² = 0.3512
    Train MAE = 0.0260, Train R² = 0.9179
    Overfit gap (R²): 0.5667
  Saved: figures/phase3/07_overfitting_diagnostic.png

  5-Fold Cross-Validation on best RF configuration...
  Paper: 'k-fold CV gives a more accurate estimate of the model's
  true predictive performance compared to evaluating on one single split'
    Fold 1: MAE = 0.0711, R² = 0.3172
    Fold 2: MAE = 0.0743, R² = 0.3700
    Fold 3: MAE = 0.0698, R² = 0.4476
    Fold 4: MAE = 0.0698, R² = 0.3629
    Fold 5: MAE = 0.0742, R² = 0.3556

    CV Mean MAE = 0.0719 ± 0.0020
    CV Mean R²  = 0.3707 ± 0.0426

======================================================================
[STEP 3.4] GRADIENT BOOSTING COMPARISON
======================================================================

  Training GradientBoostingRegressor for comparison...
  Training time: 15.9s
  Train: MAE = 0.0376, R² = 0.8875
  Val:   MAE = 0.0751, R² = 0.4021

======================================================================
[STEP 3.5] FINAL EVALUATION ON HELD-OUT TEST SET
======================================================================

  Paper: 'test data may only be used in order to evaluate a model's
  performance as a final step, after the model has been finalized.'
  'Do not evaluate your model on the held-out test data set until
  you have finished tuning your model and it is ready for publication.'

  Model comparison on VALIDATION set:
    RF (tuned):   MAE = 0.0750, R² = 0.3512
    GBR:          MAE = 0.0751, R² = 0.4021

  → Selected: Tuned Random Forest

  Re-training RandomForest (tuned) on combined train+val data...

  ╔══════════════════════════════════════════════════════╗
  ║         FINAL TEST SET RESULTS                       ║
  ╠══════════════════════════════════════════════════════╣
  ║  Model: RandomForest (tuned)                         ║
  ║  MAE  = 0.0696 eV/atom                               ║
  ║  RMSE = 0.1178 eV/atom                               ║
  ║  R²   = 0.4832                                       ║
  ║  Baseline MAE = 0.1040 eV/atom                       ║
  ║  Improvement over baseline: 33.1%                    ║
  ╚══════════════════════════════════════════════════════╝

  Saved: figures/phase3/08_final_test_parity.png
  Saved: figures/phase3/09_model_comparison.png
```

```
======================================================================
PHASE 3 COMPLETE — SUMMARY
======================================================================

  Feature Importance:
    Top features: melting_point_wmean, electron_affinity_wmean, electronegativity
_wmean, melting_point_wstd, ionization_energy_wmean
    Top property category: Melting Point

  Hyperparameter Tuning:
    Searched 108 RF configurations
    Best: n_estimators=300, max_depth=30.0,
          min_samples_split=2, min_samples_leaf=1

  Cross-Validation (5-fold):
    MAE = 0.0719 ± 0.0020
    R²  = 0.3707 ± 0.0426

  Final Test Set Performance (RandomForest (tuned)):
    MAE  = 0.0696 eV/atom
    RMSE = 0.1178 eV/atom
    R²   = 0.4832
    Improvement over baseline: 33.1%

======================================================================

>>> 11_phase3_enhanced_118.py
Working directory: D:\tco_stability_repo
======================================================================
PHASE 3 ENHANCED: 118 Magpie Features + RF / GBR / XGBoost
======================================================================

[SETUP] Loading data and applying 118-feature Magpie featurization...

  Feature breakdown (Magpie categories):
    Category 1 - Elemental property stats:  105 features
    Category 2 - Stoichiometric attributes: 7 features
    Category 3 - Electronic structure frac: 4 features
    Category 4 - Ionic compound attributes: 2 features
    ------------------------------------------
    TOTAL:                                  118 features

  Data splits: 3645 train / 1041 val / 521 test
  Total features: 118

======================================================================
[STEP 3.1] FEATURE IMPORTANCE - 118-Feature Set
======================================================================

  Top 25 most important features (with category):
  ----------------------------------------------------------------
   1. melting_point_wmean            0.0350 ##############################
   2. stoich_Lp_norm_7               0.0308 ##########################  [Stoichio
metric]
   3. stoich_Lp_norm_3               0.0297 #########################  [Stoichiom
etric]
   4. stoich_Lp_norm_10              0.0295 #########################  [Stoichiom
etric]
   5. stoich_Lp_norm_5               0.0274 ######################  [Stoichiomet
ric]
```

```
   6. stoich_fraction_max           0.0267 ######################  [Stoichiometr
ic]
   7. stoich_Lp_norm_2              0.0247 ####################  [Stoichiometri
c]
   8. electron_affinity_wmean       0.0240 ###################
   9. melting_point_wstd            0.0231 ###################
  10. ionization_energy_wmean       0.0176 ###############
  11. electronegativity_wmean       0.0174 ##############
  12. electron_affinity_wstd        0.0170 ##############
  13. electronegativity_wstd        0.0169 ##############
  14. n_valence_wstd                0.0164 ##############
  15. density_elem_wmean            0.0158 #############
  16. electronegativity_mean        0.0153 #############
  17. n_p_wmean                     0.0147 ############
  18. ionization_energy_mean        0.0147 ############
  19. group_wstd                    0.0142 ############
  20. elec_frac_p                   0.0141 ############  [Electronic Structure]
  21. row_wmean                     0.0140 ############
  22. electron_affinity_std         0.0138 ###########
  23. atomic_radius_wmean           0.0137 ###########
  24. elec_frac_s                   0.0134 ###########  [Electronic Structure]
  25. ionization_energy_wstd        0.0131 ###########

  Non-elemental feature rankings:
  ------------------------------------------------------------
    Rank   2: stoich_Lp_norm_7              (0.0308) [Stoichiometric]
    Rank   3: stoich_Lp_norm_3              (0.0297) [Stoichiometric]
    Rank   4: stoich_Lp_norm_10             (0.0295) [Stoichiometric]
    Rank   5: stoich_Lp_norm_5              (0.0274) [Stoichiometric]
    Rank   6: stoich_fraction_max           (0.0267) [Stoichiometric]
    Rank   7: stoich_Lp_norm_2              (0.0247) [Stoichiometric]
    Rank  20: elec_frac_p                   (0.0141) [Electronic Structure]
    Rank  24: elec_frac_s                   (0.0134) [Electronic Structure]
    Rank  39: elec_frac_d                   (0.0096) [Electronic Structure]
    Rank  66: ionic_max_en_diff             (0.0059) [Ionic Compound]
    Rank  79: ionic_character               (0.0046) [Ionic Compound]
    Rank  87: elec_frac_f                   (0.0037) [Electronic Structure]
    Rank  97: stoich_n_elements             (0.0022) [Stoichiometric]

  Saved: figures/phase3_enhanced/05E_feature_importance_enhanced.png
  Saved: figures/phase3_enhanced/06E_importance_by_category_enhanced.png

======================================================================
[STEP 3.2] PHYSICAL INTERPRETATION - Feature Families
======================================================================

  * Stoichiometric:
    Compositional complexity - the Lp norms capture how unevenly the
    constituent elements are distributed across the composition.
    Compositions close to a binary oxide have higher Lp norms than
    equimolar multicomponent mixtures (Ward et al., 2016).

  * Electronic Structure:
    Orbital character - the fraction of s/p/d/f valence electrons encodes
    whether the compound is dominated by main-group (s/p) or
    transition-metal (d) bonding.

  * Ionic Compound:
    Ionic character - the maximum electronegativity difference captures
    bond polarity and electrostatic stabilization in multicomponent oxides
```

```
    (Hautier et al., 2014).

======================================================================
[STEP 3.3] HYPERPARAMETER TUNING - 118 Features
======================================================================
  Total combinations: 108
  [  1/108] n=100, d=10, s=2, l=1 -> Val MAE=0.0820
  [  2/108] n=100, d=10, s=2, l=2 -> Val MAE=0.0814
  [ 36/108] n=100, d=None, s=10, l=4 -> Val MAE=0.0729
  [ 72/108] n=200, d=None, s=10, l=4 -> Val MAE=0.0728
  [108/108] n=300, d=None, s=10, l=4 -> Val MAE=0.0724

  Grid search completed in 304.9s

  BEST RF hyperparameters (by validation MAE):
    n_estimators      = 100
    max_depth         = None
    min_samples_split = 2
    min_samples_leaf  = 1
    Val MAE = 0.0685, Val R² = 0.4789
  Saved: figures/phase3_enhanced/07E_overfitting_diagnostic_enhanced.png

  5-Fold Cross-Validation on best RF and GBR configurations...
    Fold 1: RF MAE = 0.0650 (R² = 0.4329) | GBR MAE = 0.0670 (R² = 0.4507)  [Bes
t: RF]
    Fold 2: RF MAE = 0.0690 (R² = 0.4469) | GBR MAE = 0.0692 (R² = 0.4410)  [Bes
t: RF]
    Fold 3: RF MAE = 0.0649 (R² = 0.5127) | GBR MAE = 0.0641 (R² = 0.4949)  [Bes
t: GBR]
    Fold 4: RF MAE = 0.0651 (R² = 0.4450) | GBR MAE = 0.0649 (R² = 0.4031)  [Bes
t: GBR]
    Fold 5: RF MAE = 0.0685 (R² = 0.4355) | GBR MAE = 0.0698 (R² = 0.4428)  [Bes
t: RF]

    RF  CV Mean MAE = 0.0665 ± 0.0019 | Mean R² = 0.4546 ± 0.0295
    GBR CV Mean MAE = 0.0670 ± 0.0022 | Mean R² = 0.4465 ± 0.0293

======================================================================
[STEP 3.4] THREE-MODEL COMPARISON: RF vs GBR vs XGBoost
======================================================================

  Training GradientBoostingRegressor...
  GBR Training time: 17.8s
  GBR Train: MAE = 0.0344, R² = 0.9046
  GBR Val:   MAE = 0.0685, R² = 0.5057

  Training XGBRegressor...
  XGB Training time: 0.5s
  XGB Train: MAE = 0.0603, R² = 0.6897
  XGB Val:   MAE = 0.0774, R² = 0.4661

  ================================================================
  MODEL COMPARISON SUMMARY (Validation Set)
  ================================================================
  Model                          MAE         R²  Overfit Gap
  ------------------------ ---------- ---------- ------------
  Dummy (baseline)             0.1041    -0.0004          N/A
  RF (default)                 0.0700     0.4575       0.3871
  RF (tuned)                   0.0685     0.4789       0.4503
  Gradient Boosting            0.0685     0.5057       0.3989
```

```
  XGBoost                       0.0774     0.4661       0.2236
  ================================================================

======================================================================
[STEP 3.5] FINAL EVALUATION ON HELD-OUT TEST SET
======================================================================

  Validation MAE comparison:
    RF (tuned)          : 0.0685
    GBR                 : 0.0685 <- BEST
    XGBoost             : 0.0774

  Re-training GBR on combined train+val data...

  Evaluating all three models on the held-out test set...

  All models - TEST set performance (n = 521):
  Model                         MAE       RMSE         R²
  -------------------- ---------- ---------- ----------
  RF (tuned)                0.0632     0.1087     0.5596
  GBR                       0.0632     0.1029     0.6056
  XGBoost                   0.0713     0.1113     0.5384
  Saved: results/test_predictions.csv and results/final_metrics_summary.csv

  Saved: figures/phase3_enhanced/08E_final_test_parity_enhanced.png
  Saved: figures/phase3_enhanced/09E_model_comparison_enhanced.png
  Saved: figures/phase3_enhanced/10E_multi_model_parity.png

======================================================================
PHASE 3 ENHANCED - SUMMARY
======================================================================
  Features           : 118 (105 elemental + 7 stoich + 4 elec + 2 ionic)
  5-Fold CV (RF)     : MAE = 0.0665 ± 0.0019, R² = 0.4546 ± 0.0295
  5-Fold CV (GBR)    : MAE = 0.0670 ± 0.0022, R² = 0.4465 ± 0.0293
  Test Set Baseline  : MAE = 0.1040 eV/atom
  Test Set RF (tuned): MAE = 0.0632 eV/atom, RMSE = 0.1087, R² = 0.5596 (-39.2% v
s baseline)
  Test Set GBR       : MAE = 0.0632 eV/atom, RMSE = 0.1029, R² = 0.6056
  Test Set XGBoost   : MAE = 0.0713 eV/atom, RMSE = 0.1113, R² = 0.5384
======================================================================
```

Out[7]:

	model	val_mae	val_r2	test_mae	test_rmse	test_r2
0	DummyRegressor	0.104129	-0.000358	0.104020	0.163882	-0.000062
1	RF_default_118	0.069977	0.457485	NaN	NaN	NaN
2	RF_tuned_118	0.068504	0.478927	0.063181	0.108749	0.559635
3	GBR_118	0.068468	0.505713	0.063239	0.102917	0.605598
4	XGBoost_118	0.077413	0.466148	0.071307	0.111343	0.538378

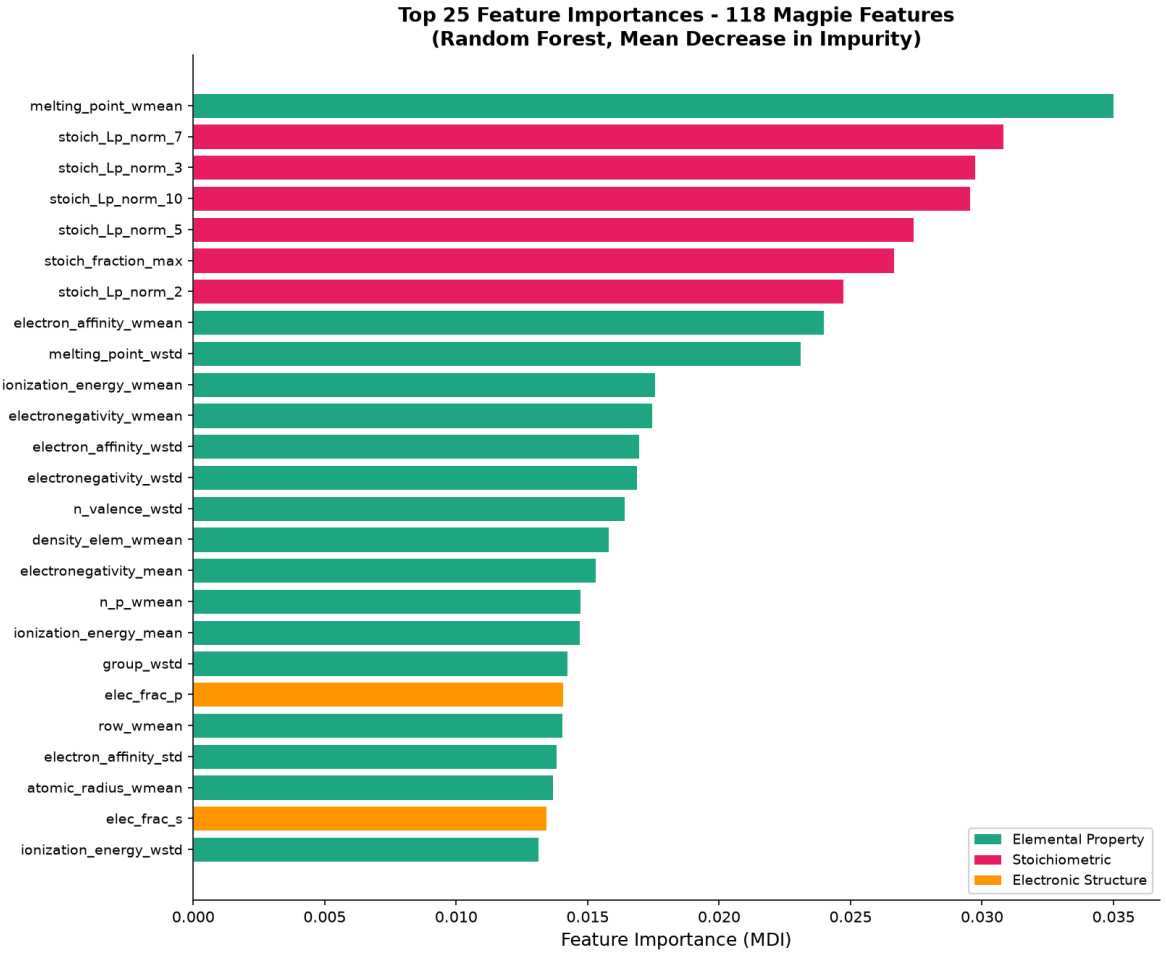
In [8]:
```
pd.read_csv(ROOT / "data/processed/grid_search_results_enhanced.csv").sort_value
```

Out[8]:

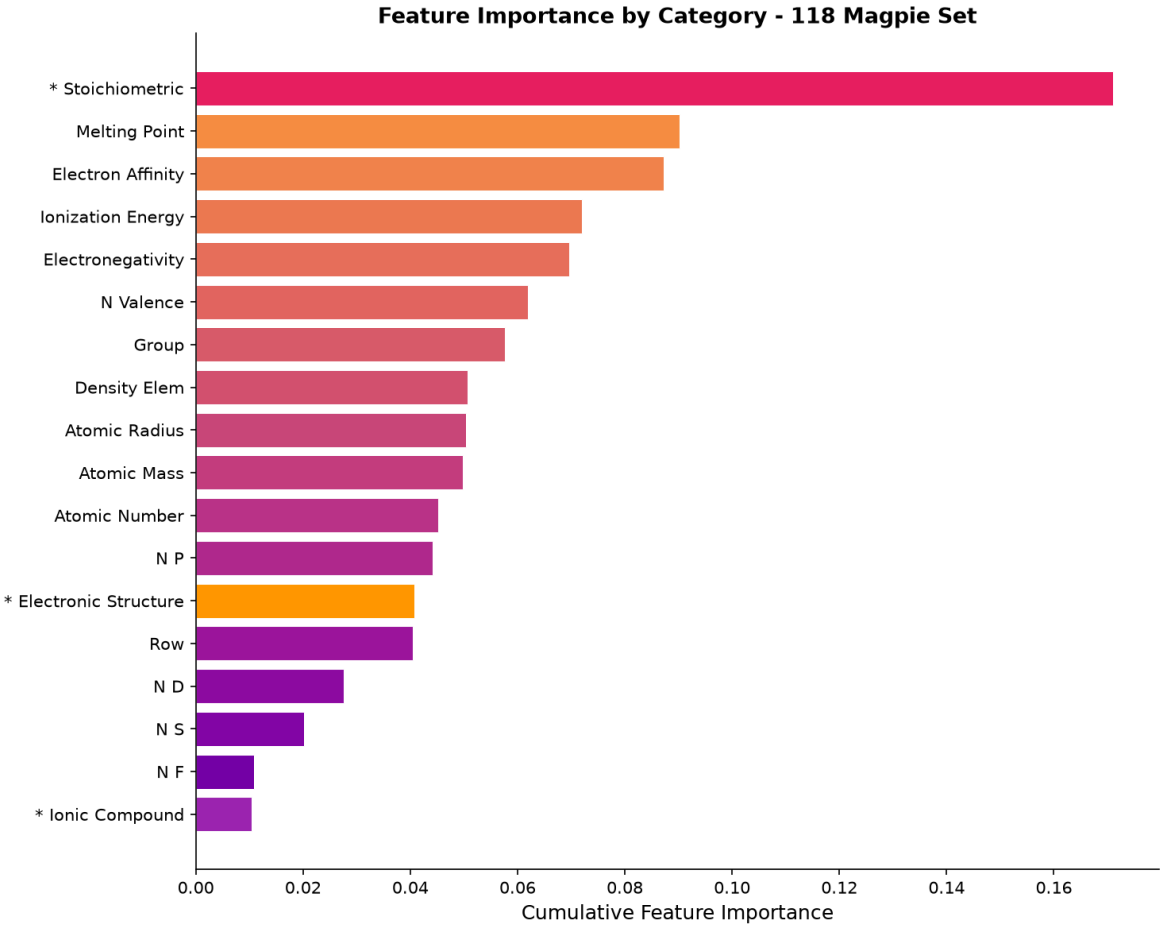
	n_estimators	max_depth	min_samples_split	min_samples_leaf	mae_train	mae_val
27	100	NaN	2	1	0.023815	0.068504
18	100	30.0	2	1	0.024279	0.068668
21	100	30.0	5	1	0.030314	0.068946
99	300	NaN	2	1	0.023717	0.069096
90	300	30.0	2	1	0.024175	0.069147
54	200	30.0	2	1	0.024231	0.069173
30	100	NaN	5	1	0.030137	0.069429
63	200	NaN	2	1	0.023825	0.069469
93	300	30.0	5	1	0.030172	0.069488
57	200	30.0	5	1	0.030367	0.069490

In [9]:
```
show("figures/phase3_enhanced/*.png")
```

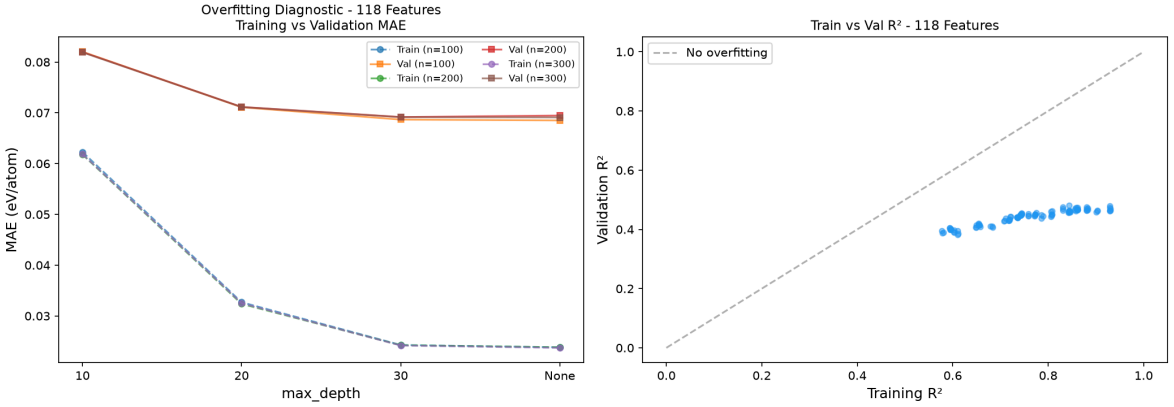
figures\phase3_enhanced\05E_feature_importance_enhanced.png



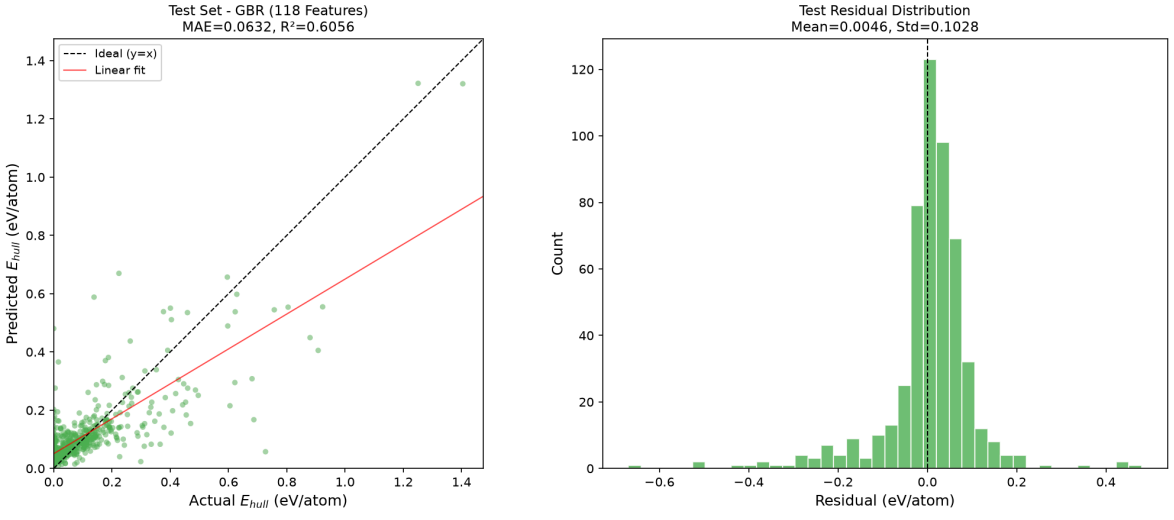
figures\phase3_enhanced\06E_importance_by_category_enhanced.png



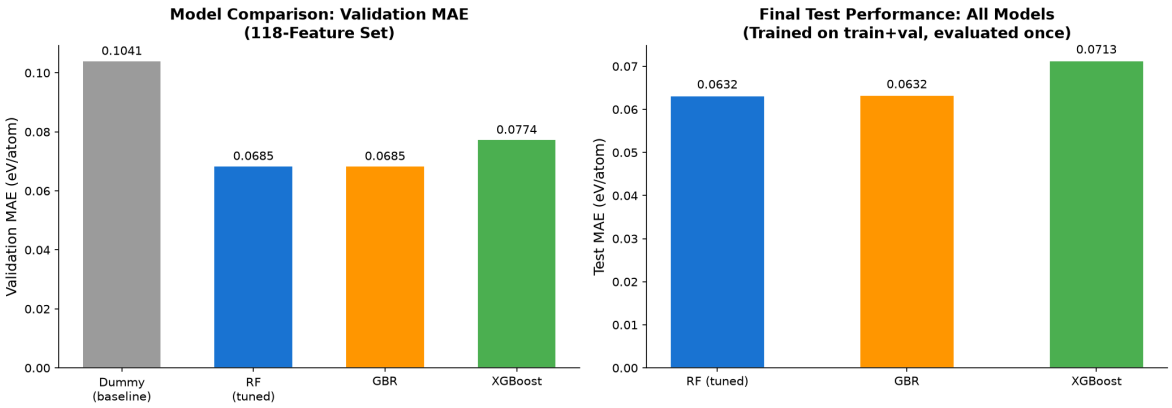
figures\phase3_enhanced\07E_overfitting_diagnostic_enhanced.png



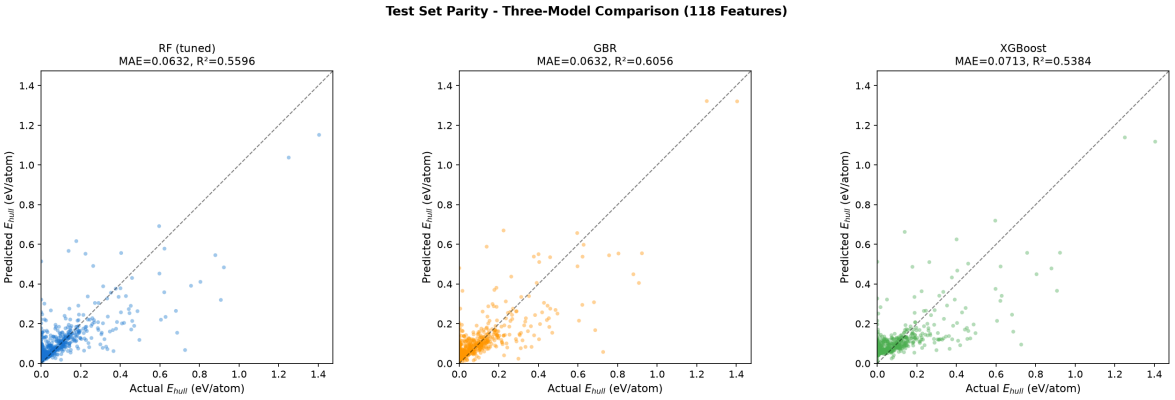
figures\phase3_enhanced\08E_final_test_parity_enhanced.png



figures\phase3_enhanced\09E_model_comparison_enhanced.png



figures\phase3_enhanced\10E_multi_model_parity.png



Publication Figures, Validation PFI, LOHO, and Dopant Sensitivity (Sections 3.6-3.8)

In [10]:
```
for s in ["12_pub_figures.py", "13_permutation_importance.py", "14_leave_one_hos
          "15_dopant_sensitivity.py", "16_synthetic_inference.py"]:
    run(s)
```

```
>>> 12_pub_figures.py
Working directory: D:\tco_stability_repo
Loading data...
  5207 samples, 118 features
  Split: 3645 train / 1041 val / 521 test
Training models...
  XGBoost: available
  RF:  MAE=0.0632, R2=0.5596, RMSE=0.1087
  GBR: MAE=0.0632
  XGB: MAE=0.0713

Fig 1: Dataset overview...
  Saved fig1
Fig 2: Baseline parity...
  Saved fig2
Fig 3: Feature importance...
  Saved fig3
Fig 4: Category importance...
  Saved fig4
Fig 5: Model comparison...
  Saved fig5
Fig 6: Final test parity + residuals...
  Saved fig6

============================================================
All 6 publication figures saved to: figures/publication/
  fig1_dataset_overview          (.png + .pdf)
  fig2_baseline_parity           (.png + .pdf)
  fig3_feature_importance        (.png + .pdf)
  fig4_category_importance       (.png + .pdf)
  fig5_model_comparison          (.png + .pdf)
  fig6_final_test_results        (.png + .pdf)
============================================================

>>> 13_permutation_importance.py
Working directory: D:\tco_stability_repo
======================================================================
STEP 13: Permutation Feature Importance Analysis (Validation Set)
======================================================================

[1/5] Loading and featurizing data...
  5207 samples, 118 features

[2/5] Splitting data (composition-based, seed=42)...
  Train: 3645 | Val: 1041 | Test: 521

[3/5] Training tuned RF on training set only (preserving validation & test integr
ity)...
  Validation MAE = 0.0695 eV/atom, R² = 0.4640 (Test set untouched)

[4/5] Computing MDI and Validation-Set Permutation (PFI) importances...
  Permutation importance computed in 204.1s (30 repeats × 118 features)
  Full results saved to results/permutation_importance_full.csv

======================================================================
KEY COMPARISON: MDI vs PFI Top 10 (Validation Set)
======================================================================

  MDI Top 10:
     1. melting_point_wmean                  0.0310  [ELEM]
```

```
     2. stoich_Lp_norm_7                    0.0284  [STOICH]
     3. stoich_Lp_norm_3                    0.0279  [STOICH]
     4. stoich_Lp_norm_2                    0.0268  [STOICH]
     5. melting_point_wstd                  0.0267  [ELEM]
     6. stoich_Lp_norm_10                   0.0264  [STOICH]
     7. stoich_Lp_norm_5                    0.0259  [STOICH]
     8. electron_affinity_wmean             0.0251  [ELEM]
     9. stoich_fraction_max                 0.0223  [STOICH]
    10. electronegativity_wmean             0.0177  [ELEM]

  PFI Top 10 (unbiased, on validation set):
     1. stoich_Lp_norm_3                    0.0056 ± 0.0004  [STOICH]
     2. stoich_Lp_norm_10                   0.0050 ± 0.0003  [STOICH]
     3. stoich_Lp_norm_7                    0.0049 ± 0.0003  [STOICH]
     4. stoich_Lp_norm_5                    0.0049 ± 0.0003  [STOICH]
     5. stoich_Lp_norm_2                    0.0042 ± 0.0003  [STOICH]
     6. stoich_fraction_max                 0.0041 ± 0.0003  [STOICH]
     7. ionization_energy_min               0.0034 ± 0.0003  [ELEM]
     8. electronegativity_wstd              0.0034 ± 0.0003  [ELEM]
     9. ionization_energy_range             0.0033 ± 0.0002  [ELEM]
    10. melting_point_wmean                 0.0033 ± 0.0003  [ELEM]

  Stoichiometric features in MDI top 10: 6
  Stoichiometric features in PFI top 10: 6

[5/5] Generating figures...
  Saved: figures/publication/fig_mdi_vs_pfi_comparison.{png,pdf}

======================================================================
PERMUTATION IMPORTANCE ANALYSIS COMPLETE
======================================================================

>>> 14_leave_one_host_out.py
Working directory: D:\tco_stability_repo
============================================================
STEP 14: LEAVE-ONE-HOST-OUT VALIDATION (Section 3.7)
============================================================

Loading data...
Featurizing...
  5207 samples, 118 features
  Zn: 2605, In: 946, Sn: 1656

  Hold out: Zn-oxides
    Train: 2602 samples (In, Sn)
    Test:  2605 samples (Zn)
    MAE  = 0.1097 eV/atom
    RMSE = 0.1726 eV/atom
    R²   = 0.1274
    Baseline MAE = 0.1053
    Improvement  = -4.2%

  Hold out: In-oxides
    Train: 4261 samples (Zn, Sn)
    Test:  946 samples (In)
    MAE  = 0.0795 eV/atom
    RMSE = 0.1295 eV/atom
    R²   = 0.1282
    Baseline MAE = 0.0998
    Improvement  = 20.3%
```

```
  Hold out: Sn-oxides
    Train: 3551 samples (Zn, In)
    Test:  1656 samples (Sn)
    MAE  = 0.0728 eV/atom
    RMSE = 0.1215 eV/atom
    R²   = 0.3210
    Baseline MAE = 0.0948
    Improvement  = 23.3%

======================================================================
COMPARISON: Leave-One-Host-Out vs Standard Composition Split
======================================================================
  Standard split (composition-based):
    Test MAE = 0.0632 eV/atom, R² = 0.5596 (RF) / 0.6056 (GBR)

  Leave-one-host-out:
    Hold out Zn: MAE = 0.1097, R² = 0.1274
    Hold out In: MAE = 0.0795, R² = 0.1282
    Hold out Sn: MAE = 0.0728, R² = 0.3210

    Average:     MAE = 0.0873, R² = 0.1922

Generating Figure S1 (Manuscript Fig. 8)...
  Saved: figures/publication/figS1_leave_one_host_out.{png,pdf}
Generating Figure S2 (Manuscript Fig. 9)...
  Saved: figures/publication/figS2_loho_comparison.{png,pdf}

======================================================================
LEAVE-ONE-HOST-OUT VALIDATION COMPLETE
======================================================================

>>> 15_dopant_sensitivity.py
Working directory: D:\tco_stability_repo
======================================================================
STEP 15: DOPANT SENSITIVITY ANALYSIS (Section 3.8)
======================================================================

Computing feature vectors across dopant concentrations...

  ZnO:Al (AZO) at 1% dopant:
    L2 distance:             392.0282
    Stoichiometric shift:    1.0000
    Weighted-mean shift:     1.2302
    Max relative change:     0.8068 (80.7%)

  ZnO:Al (AZO) at 5% dopant:
    L2 distance:             392.1557
    Stoichiometric shift:    1.0005
    Weighted-mean shift:     6.1510
    Max relative change:     0.8068 (80.7%)

  ZnO:Al (AZO) at 10% dopant:
    L2 distance:             392.5417
    Stoichiometric shift:    1.0018
    Weighted-mean shift:     12.3019
    Max relative change:     0.8068 (80.7%)

  ZnO:Ga (GZO) at 1% dopant:
    L2 distance:             64.4422
```

```
  Stoichiometric shift:    1.0000
  Weighted-mean shift:     1.9498
  Max relative change:     0.8037 (80.4%)

ZnO:Ga (GZO) at 5% dopant:
  L2 distance:             65.2631
  Stoichiometric shift:    1.0005
  Weighted-mean shift:     9.7491
  Max relative change:     0.8037 (80.4%)

ZnO:Ga (GZO) at 10% dopant:
  L2 distance:             67.8057
  Stoichiometric shift:    1.0018
  Weighted-mean shift:     19.4983
  Max relative change:     0.8037 (80.4%)

In$_2$O$_3$:Sn (ITO) at 1% dopant:
  L2 distance:             140.6107
  Stoichiometric shift:    1.0000
  Weighted-mean shift:     0.3051
  Max relative change:     0.5000 (50.0%)

In$_2$O$_3$:Sn (ITO) at 5% dopant:
  L2 distance:             140.6345
  Stoichiometric shift:    1.0001
  Weighted-mean shift:     1.5255
  Max relative change:     0.5000 (50.0%)

In$_2$O$_3$:Sn (ITO) at 10% dopant:
  L2 distance:             140.7076
  Stoichiometric shift:    1.0003
  Weighted-mean shift:     3.0510
  Max relative change:     0.5000 (50.0%)

SnO$_2$:F (FTO) at 1% dopant:
  L2 distance:             81.4339
  Stoichiometric shift:    1.0000
  Weighted-mean shift:     1.5862
  Max relative change:     5.5587 (555.9%)

SnO$_2$:F (FTO) at 5% dopant:
  L2 distance:             81.8548
  Stoichiometric shift:    1.0000
  Weighted-mean shift:     7.9308
  Max relative change:     5.5587 (555.9%)

SnO$_2$:F (FTO) at 10% dopant:
  L2 distance:             83.1795
  Stoichiometric shift:    1.0001
  Weighted-mean shift:     15.8615
  Max relative change:     5.5587 (555.9%)

SnO$_2$:Sb (ATO) at 1% dopant:
  L2 distance:             613.6714
  Stoichiometric shift:    1.0000
  Weighted-mean shift:     1.3297
  Max relative change:     0.8846 (88.5%)

SnO$_2$:Sb (ATO) at 5% dopant:
  L2 distance:             613.8814
```

```
    Stoichiometric shift:    1.0000
    Weighted-mean shift:     6.6483
    Max relative change:     0.8846 (88.5%)

  SnO$_2$:Sb (ATO) at 10% dopant:
    L2 distance:             614.4857
    Stoichiometric shift:    1.0001
    Weighted-mean shift:     13.2967
    Max relative change:     0.8846 (88.5%)

================================================================
DETAILED: Feature shifts for 1% Al in ZnO
================================================================

  Feature                                Undoped      1% Al       Abs Δ     Rel Δ
  ------------------------------------------------------------------------------
  electron_affinity_min                   0.0000     0.0000     0.0000       new
[Elemental]
  n_s_std                                 0.0000     0.0000     0.0000       new
[Elemental]
  n_s_range                               0.0000     0.0000     0.0000       new
[Elemental]
  n_s_wstd                                0.0000     0.0000     0.0000       new
[Elemental]
  n_p_min                                 0.0000     0.0000     0.0000       new
[Elemental]
  n_d_min                                 0.0000     0.0000     0.0000       new
[Elemental]
  n_f_mean                                0.0000     0.0000     0.0000       new
[Elemental]
  n_f_std                                 0.0000     0.0000     0.0000       new
[Elemental]
  n_f_min                                 0.0000     0.0000     0.0000       new
[Elemental]
  n_f_max                                 0.0000     0.0000     0.0000       new
[Elemental]
  n_f_range                               0.0000     0.0000     0.0000       new
[Elemental]
  n_f_wmean                               0.0000     0.0000     0.0000       new
[Elemental]
  n_f_wstd                                0.0000     0.0000     0.0000       new
[Elemental]
  elec_frac_f                             0.0000     0.0000     0.0000       new
[Electronic]
  ionization_energy_range                 4.2240     7.6320     3.4080     80.7%
[Elemental]
  n_valence_min                           6.0000     3.0000     3.0000     50.0%
[Elemental]
  n_valence_range                         6.0000     9.0000     3.0000     50.0%
[Elemental]
  stoich_n_elements                       2.0000     3.0000     1.0000     50.0%
[Stoichiometric]
  melting_point_mean                    373.5200   560.1700   186.6500     50.0%
[Elemental]
  ionization_energy_std                   2.1120     3.1217     1.0097     47.8%
[Elemental]

  Features changing <1%:  66/118 (56%)
  Features changing <5%:  74/118 (63%)
```

```
================================================================
Generating Figure S3 (Manuscript Fig. 10)...
findfont: Failed to find font weight semibold for Times New Roman, now using 700.
  Saved: figures/publication/figS3_dopant_sensitivity.{png,pdf}

================================================================
DOPANT SENSITIVITY ANALYSIS COMPLETE
================================================================

>>> 16_synthetic_inference.py
Working directory: D:\tco_stability_repo
======================================================================
STEP 16: Synthetic Dopant Inference Test (Section 3.8)
======================================================================

[1/4] Rebuilding pipeline and training model...
  Model trained on 4686 samples, 118 features

[2/4] Generating synthetic doped compositions...
  Generated 852 synthetic predictions
  Saved to results/synthetic_inference_predictions.csv

[3/4] Analyzing concentration sensitivity of predictions...

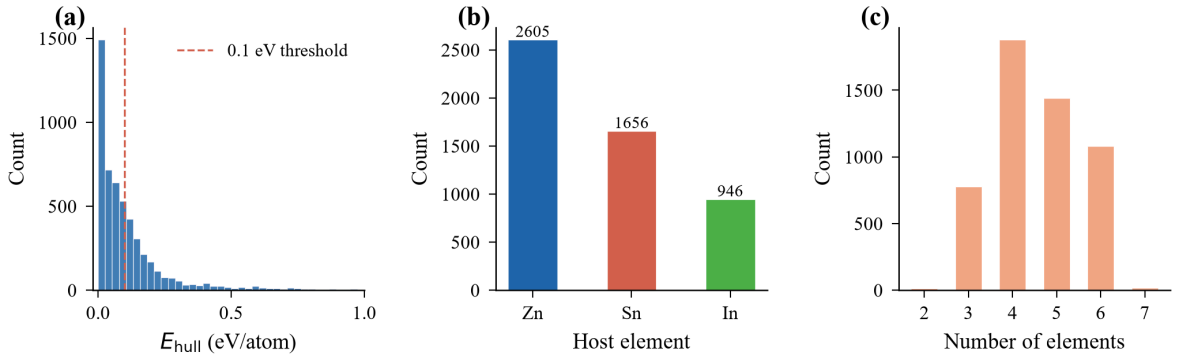
  Predicted Ehull at key concentrations for ZnO:Al:
      0.5% Al -> predicted Ehull = 0.1410 eV/atom
      1.0% Al -> predicted Ehull = 0.1508 eV/atom
      2.0% Al -> predicted Ehull = 0.1516 eV/atom
      3.0% Al -> predicted Ehull = 0.1493 eV/atom
      5.0% Al -> predicted Ehull = 0.1325 eV/atom
     10.0% Al -> predicted Ehull = 0.1204 eV/atom
     20.0% Al -> predicted Ehull = 0.1139 eV/atom
     50.0% Al -> predicted Ehull = 0.1496 eV/atom
  ZnO:Al  1-5% range: 0.0227 eV/atom | 5-50% range: 0.0565 eV/atom
  ZnO:Ga  1-5% range: 0.0183 eV/atom | 5-50% range: 0.0336 eV/atom
  ZnO:F   1-5% range: 0.0126 eV/atom | 5-50% range: 0.0813 eV/atom
  ZnO:Sb  1-5% range: 0.0253 eV/atom | 5-50% range: 0.0590 eV/atom
  In2O3:Al  1-5% range: 0.0082 eV/atom | 5-50% range: 0.0263 eV/atom
  In2O3:Ga  1-5% range: 0.0032 eV/atom | 5-50% range: 0.0185 eV/atom
  In2O3:F   1-5% range: 0.0048 eV/atom | 5-50% range: 0.0959 eV/atom
  In2O3:Sb  1-5% range: 0.0132 eV/atom | 5-50% range: 0.2043 eV/atom
  SnO2:Al  1-5% range: 0.0057 eV/atom | 5-50% range: 0.1191 eV/atom
  SnO2:Ga  1-5% range: 0.0069 eV/atom | 5-50% range: 0.0654 eV/atom
  SnO2:F   1-5% range: 0.0048 eV/atom | 5-50% range: 0.0405 eV/atom
  SnO2:Sb  1-5% range: 0.0043 eV/atom | 5-50% range: 0.0157 eV/atom

[4/4] Generating figures...
The PostScript backend does not support transparency; partially transparent artis
ts will be rendered opaque.
  Saved: figures/publication/fig_synthetic_inference_all_hosts.{png,pdf}

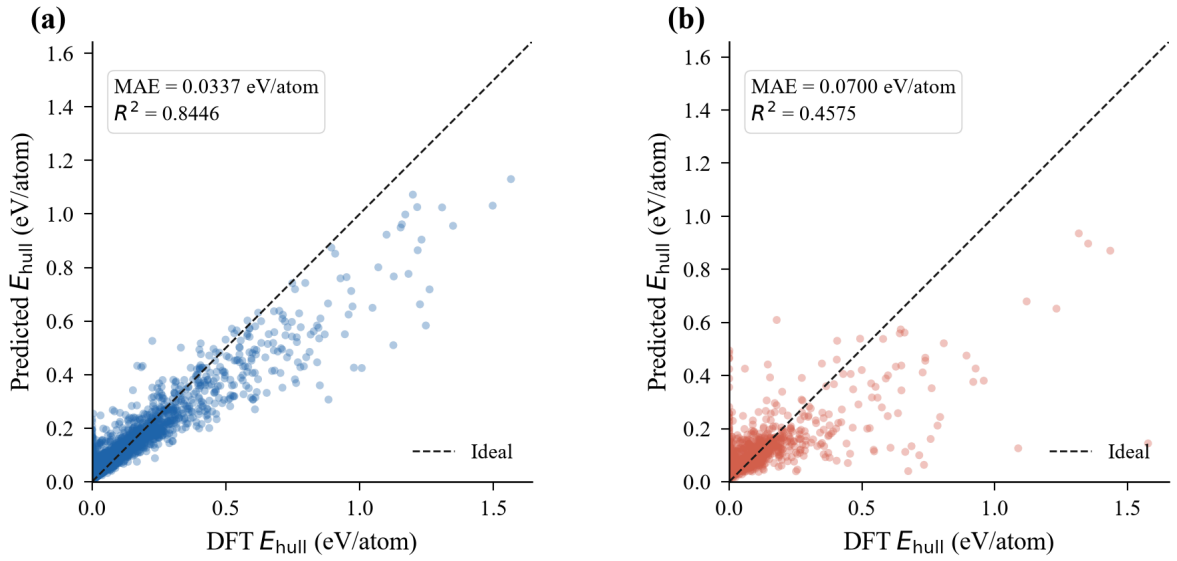
======================================================================
SYNTHETIC INFERENCE TEST COMPLETE
======================================================================
```

In [11]:
```
show("figures/publication/fig*.png")
```

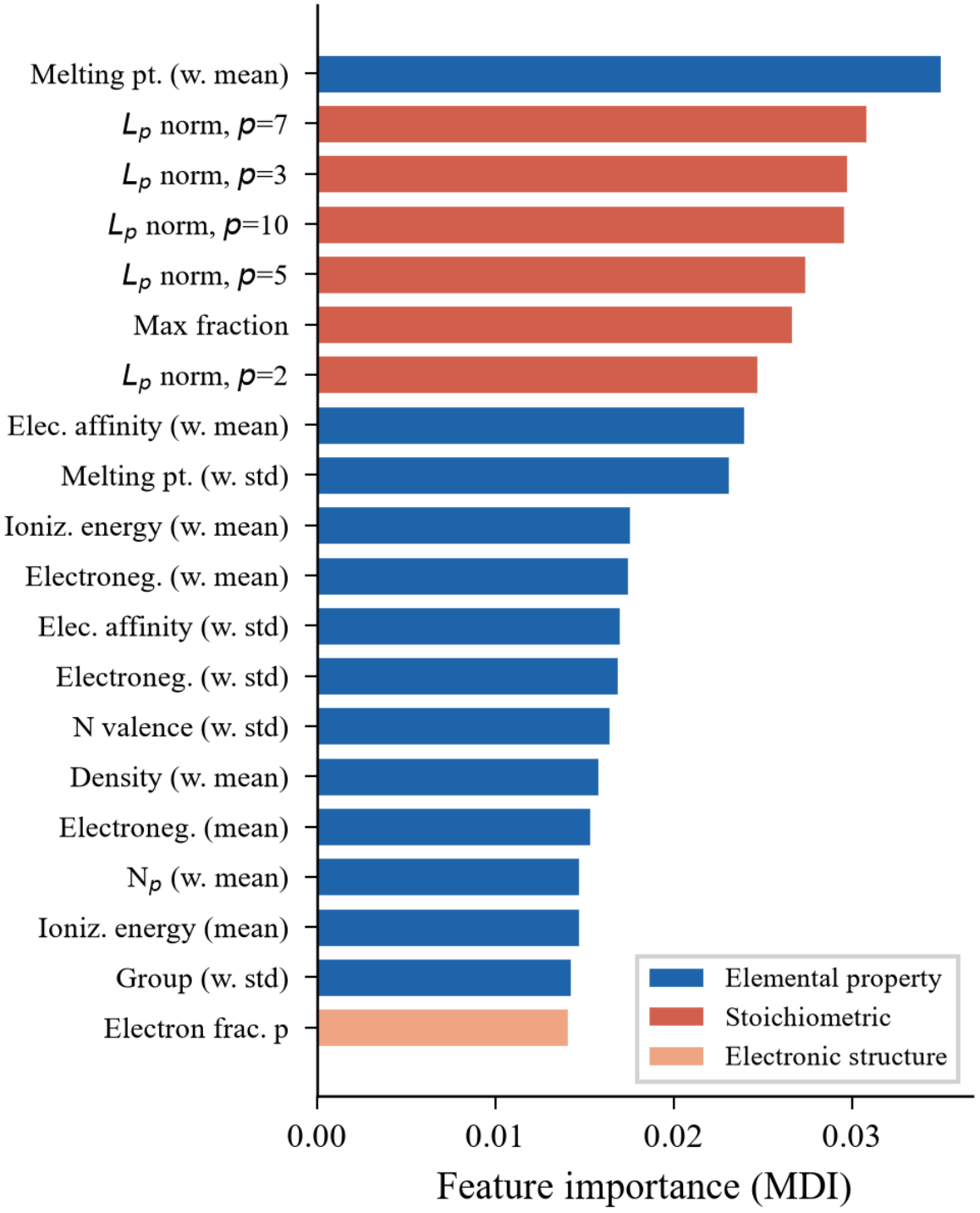
```
figures\publication\fig1_dataset_overview.png
```



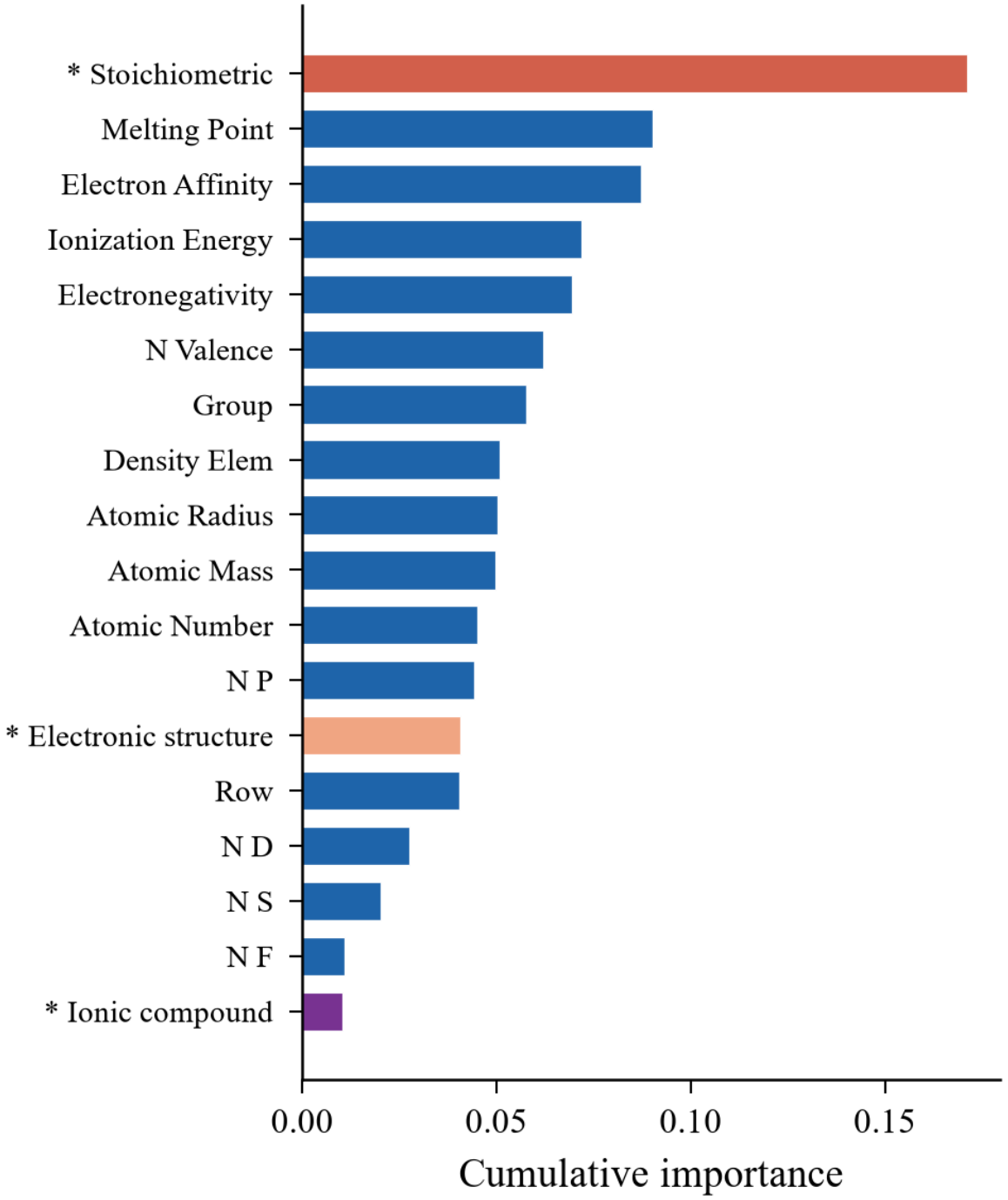
figures\publication\fig2_baseline_parity.png



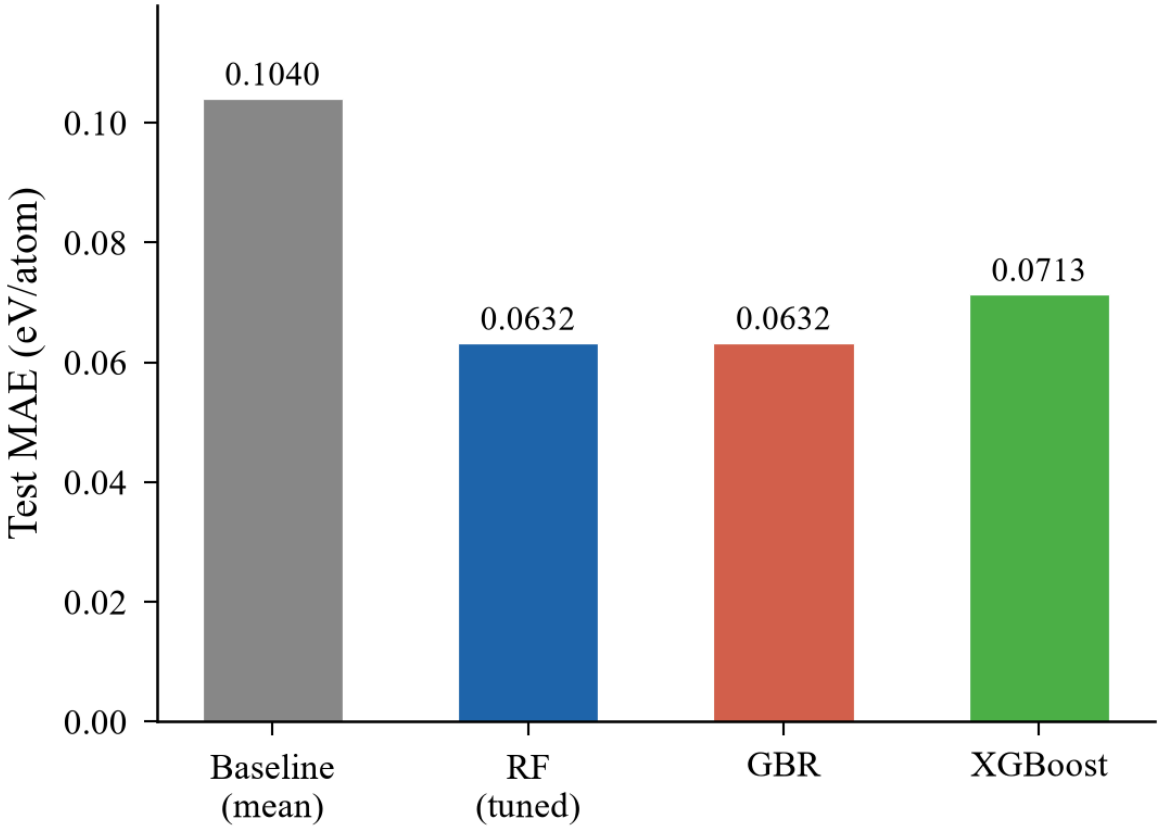
figures\publication\fig3_feature_importance.png



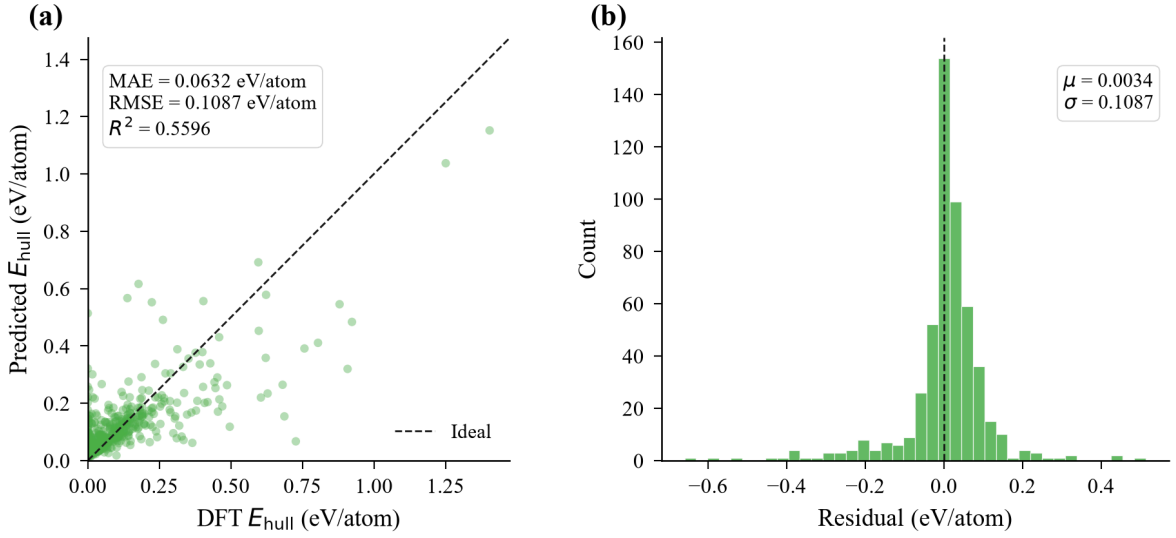
figures\publication\fig4_category_importance.png



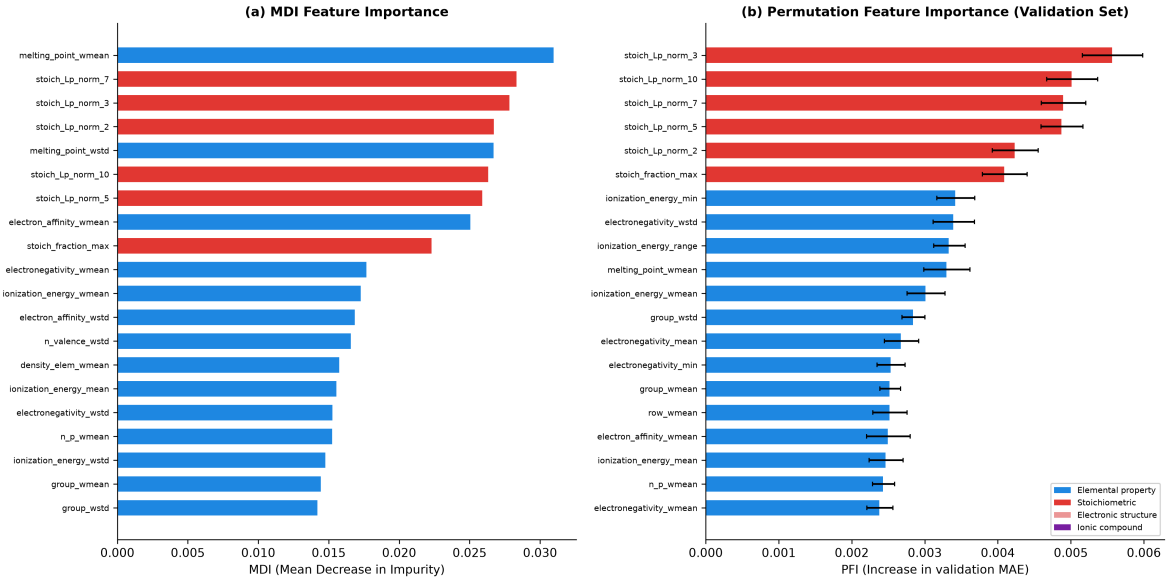
figures\publication\fig5_model_comparison.png



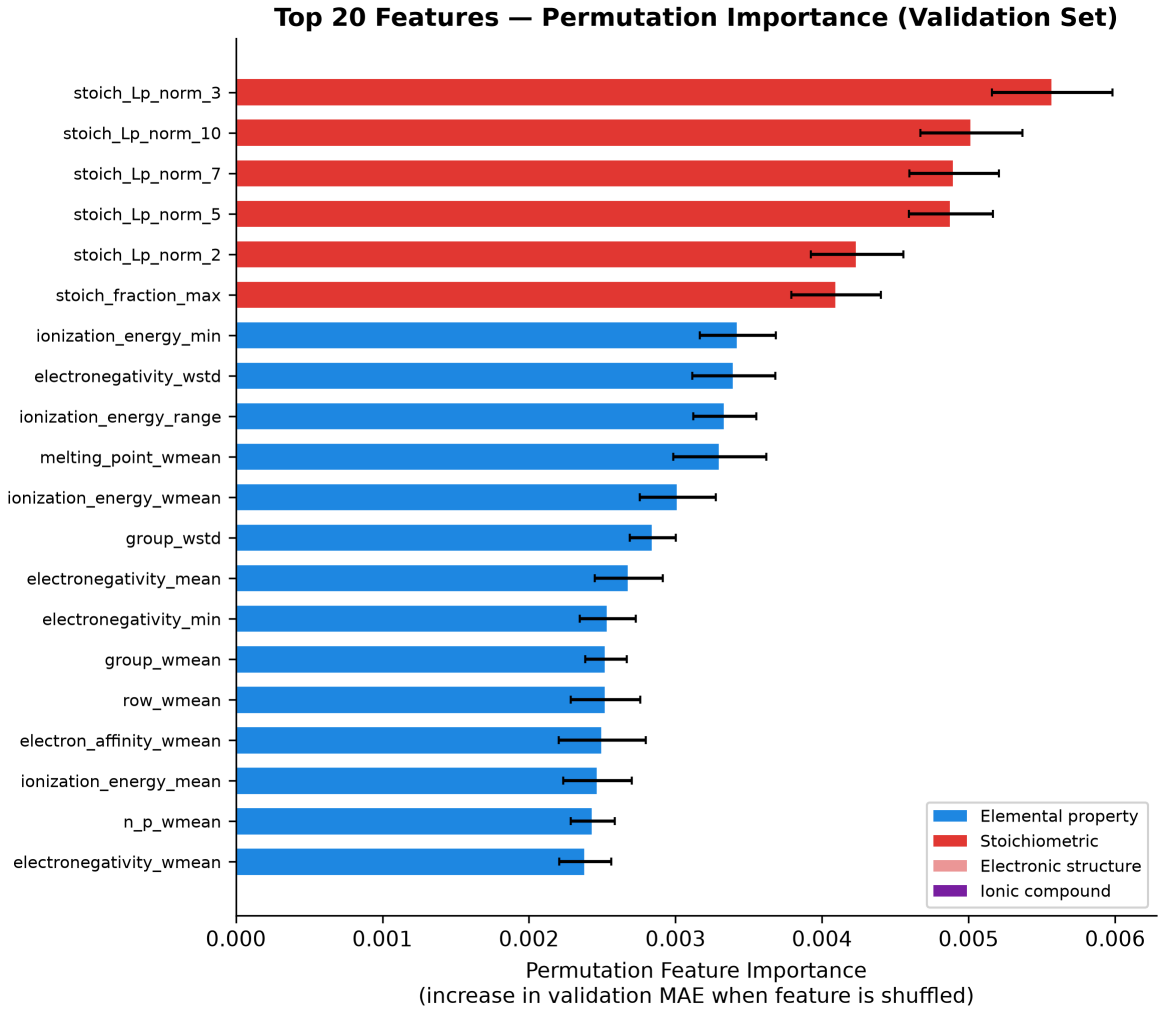
figures\publication\fig6_final_test_results.png



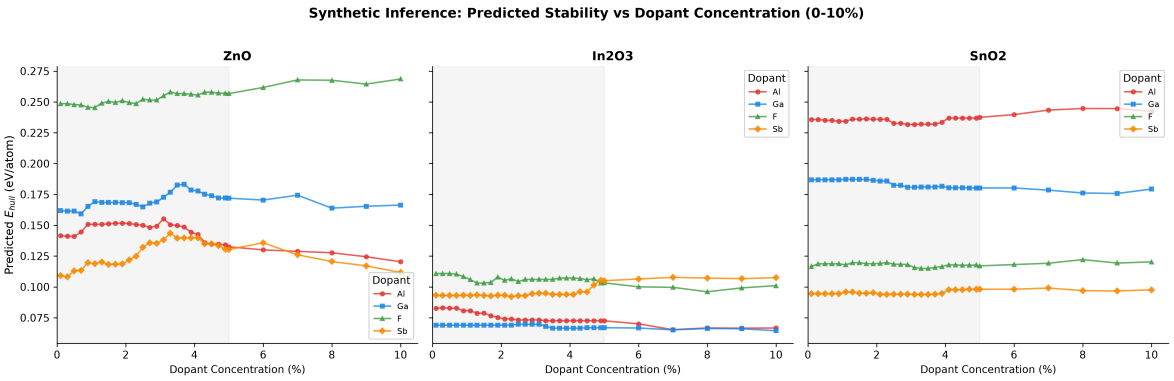
figures\publication\fig_mdi_vs_pfi_comparison.png



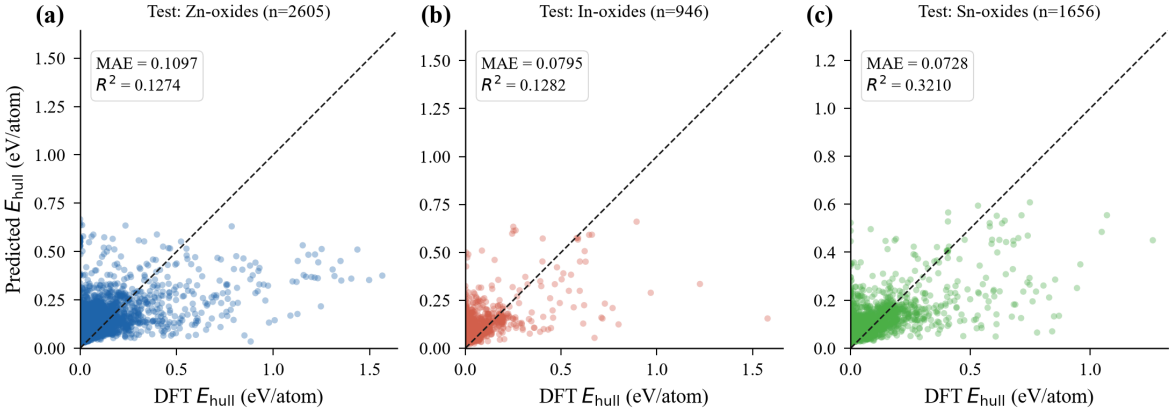
figures\publication\fig_pfi_top20.png



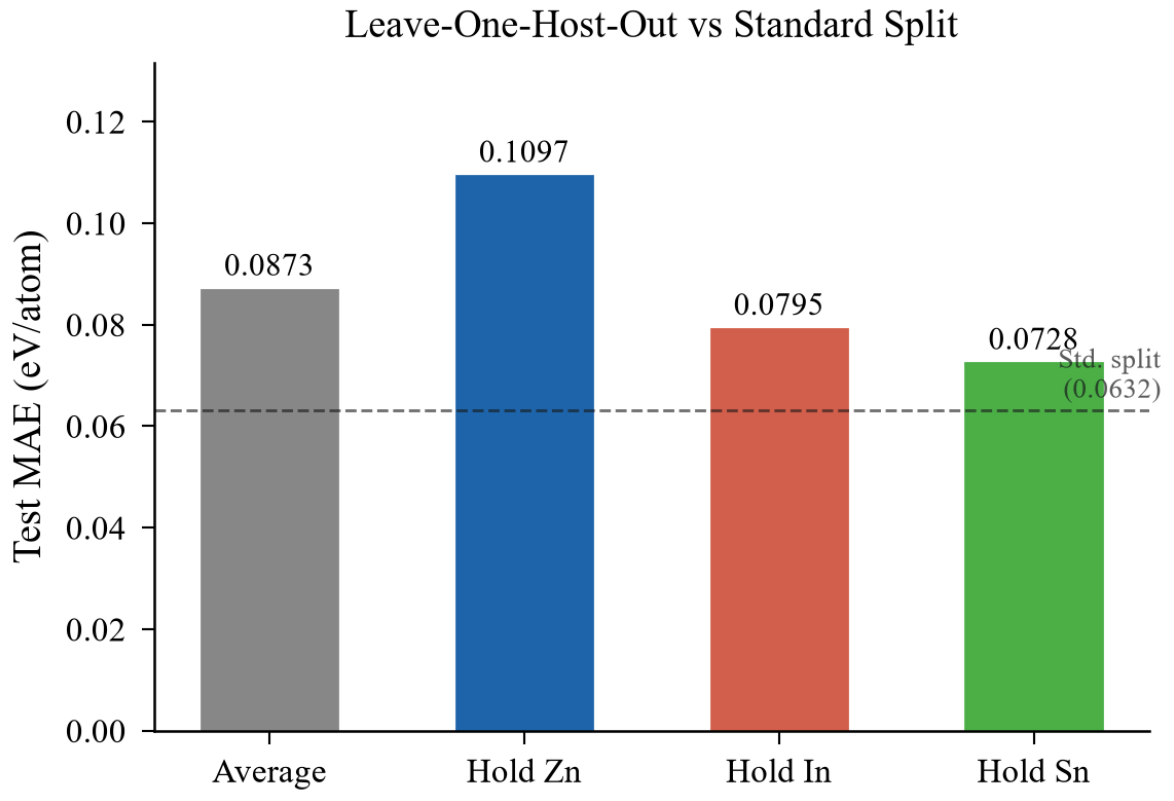
figures\publication\fig_synthetic_inference_all_hosts.png



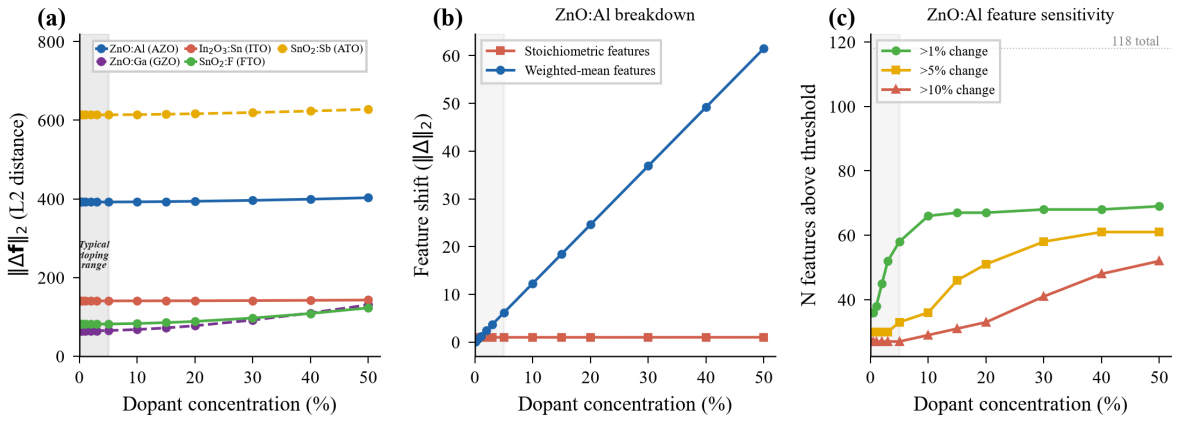
figures\publication\figS1_leave_one_host_out.png



figures\publication\figS2_loho_comparison.png



figures\publication\figS3_dopant_sensitivity.png



Export Trained Models

In [12]:
```
run("17_save_models.py")
print(sorted(p.name for p in (ROOT / "models").glob("*")))
```

```
>>> 17_save_models.py
Working directory: D:\tco_stability_repo
[1/3] Rebuilding the 118-feature pipeline...
  3645 train / 1041 val / 521 test, 118 features

[2/3] Reading the best RF hyperparameters from the grid search...
  {'n_estimators': 100, 'max_depth': None, 'min_samples_split': 2, 'min_samples_l
eaf': 1, 'max_features': 'sqrt', 'random_state': 42, 'n_jobs': -1}

[3/3] Training final models on train+val and saving...
  random_forest      test MAE = 0.0632, R² = 0.5596  -> models/random_forest.jobl
ib
  gradient_boosting  test MAE = 0.0632, R² = 0.6056  -> models/gradient_boosting.
joblib
  xgboost            test MAE = 0.0713, R² = 0.5384  -> models/xgboost.joblib

  Saved scaler, feature names and summary to models/
['.gitkeep', 'feature_names.joblib', 'gradient_boosting.joblib', 'model_summary.j
son', 'random_forest.joblib', 'scaler.joblib', 'xgboost.joblib']
```

Verification Against Reference Outputs

Compares regenerated CSV tables with those in `reference/` (tolerance `1e-9`).

In [13]:
```
run("verify_outputs.py")
```

```
>>> verify_outputs.py
Working directory: D:\tco_stability_repo
======================================================================
STRUCTURAL CHECKS
======================================================================
  [OK] cleaned dataset rows (5216 vs 5216)
  [OK] train split size (3645 vs 3645)
  [OK] val split size (1041 vs 1041)
  [OK] test split size (521 vs 521)

======================================================================
NUMERICAL COMPARISON WITH reference/
======================================================================
  [OK] ambiguous_cases.csv
  [OK] feature_importances_enhanced.csv
  [OK] grid_search_results_enhanced.csv
  [OK] phase2_model_results.csv
  [OK] test.csv
  [OK] train.csv
  [OK] val.csv

======================================================================
ALL CHECKS PASSED
```